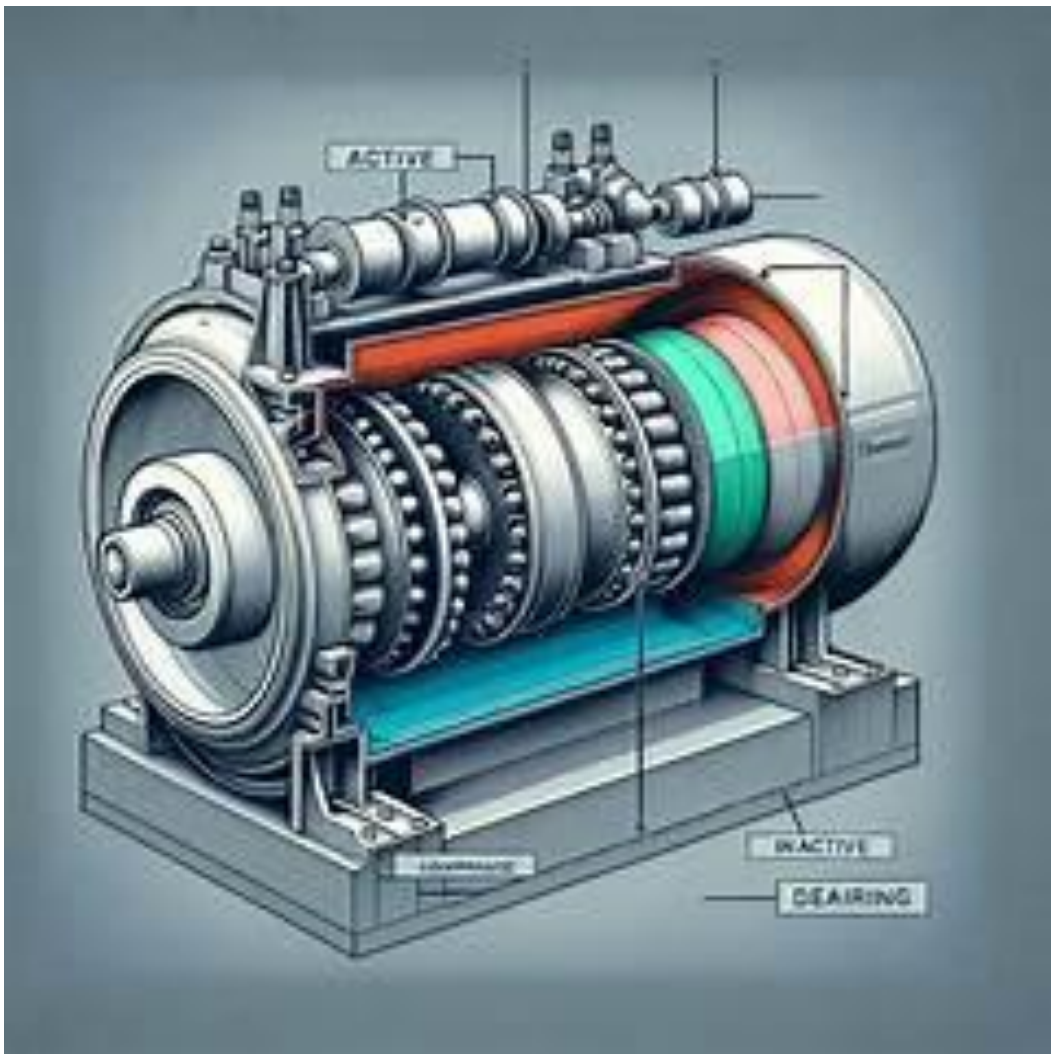
Exp 2

DC generator



Name: Abdelrahman sherif

Code:2300100 sec: 5

Spring 25

DC Motor construction:

 Armature: The rotating part where EMF is induced.

 Field Magnets: Provide the magnetic field.

 Commutator: Converts the alternating current (AC) induced in the armature into direct current (DC).

 Brushes: Conduct the current from the commutator to the external circuit.

Working Principle:

 Magnetic Field Creation: The field magnets create a magnetic field in the air gap between the stator and the rotor (armature).

 Armature Rotation: The armature, which consists of a coil, is rotated mechanically using an external force (like a prime mover).

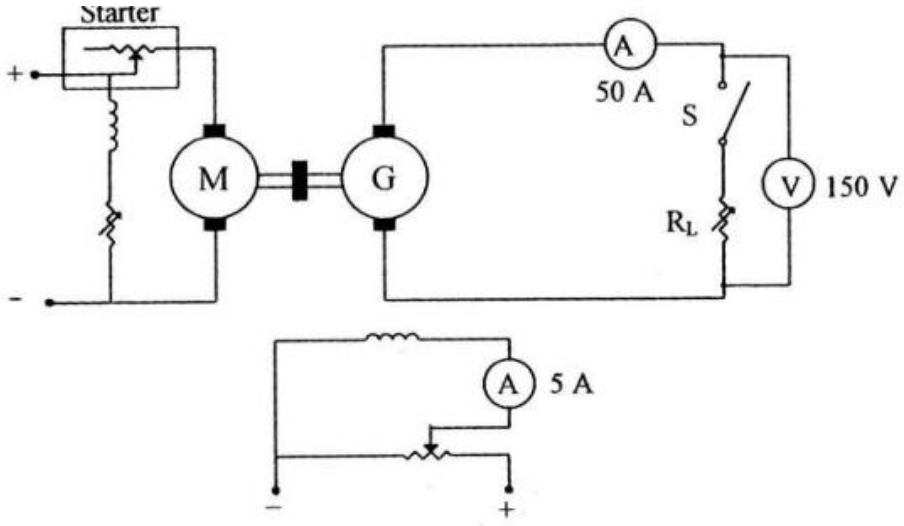
 EMF Induction: As the armature coil rotates, it cuts the magnetic flux, and an EMF is induced according to Faraday's law. The direction of the induced EMF is given by the Fleming's right-hand rule.

 Current Collection: The induced EMF generates an alternating current (AC) in the armature windings.

 Commutation: The commutator, which is segmented and connected to the armature, rectifies the AC into DC. It ensures that the current flowing in the external circuit always flows in one direction.

 Current Flow: Brushes, which are in contact with the commutator, collect the DC and transfer it to the external circuit.

Open circuit characteristics



 Circuit Setup: Assemble the circuit as illustrated, ensuring all connections are secure.

 Initial Operation: Keep the armature terminals open (with switch S open-circuited) and rotate the rotor at a constant rated speed. Adjust the field current of the coupled DC motor to maintain this speed.

 Field Current Variation: Gradually increase the generator's field current from zero until magnetic saturation is achieved, while recording the corresponding values of field current (If) and generated voltage (V).

 Plotting the OCC: Create the Open Circuit Characteristic (OCC) curve by plotting the generated voltage (V) against the field current (If).

 Reduced Speed Testing: Repeat steps 3 and 4, but this time adjust the rotor speed to 80% of its rated value and plot the corresponding OCC.

During Load test explain what will happen to the speed of the motor driving the generator?

When the load on the generator increases, more current flows through the armature, **causing:**

1-Increased armature reaction, which weakens the magnetic field.

2-Increased voltage drop due to armature resistance.

3-Higher mechanical power demand from the prime mover (motor).

Lab readings

I-load	10A	20A	30A	40A
V-load	11.8	11.1V	10.4V	9.9V

